

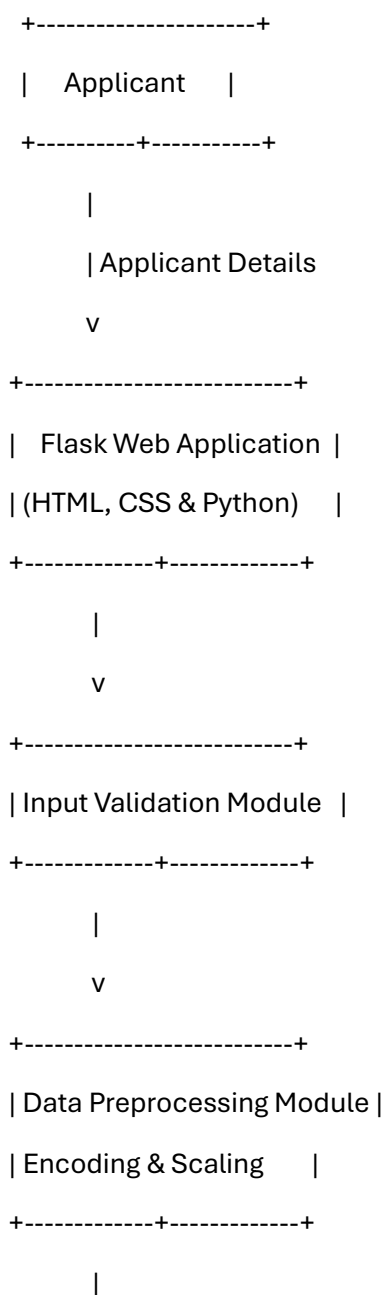
1. Introduction

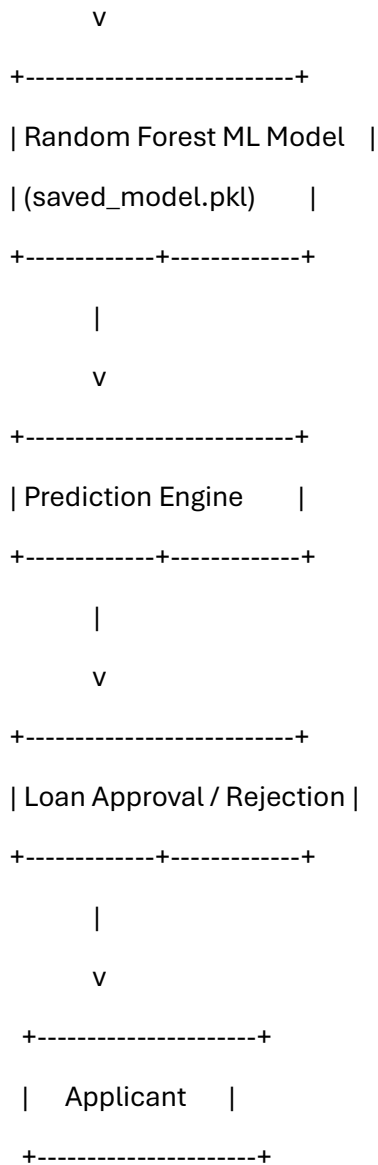
The **Smart Lender** system is a Machine Learning-based web application designed to automate the loan eligibility prediction process. It allows users to enter applicant details through a web interface, processes the data using a trained machine learning model, and instantly predicts whether the loan application is likely to be approved or rejected.

The architecture follows a modular approach, making the system scalable, maintainable, and easy to enhance in the future.

2. Solution Architecture Diagram

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3. Architecture Components

Component	Description
User Interface	Built using HTML and CSS to collect applicant information.
Flask Backend	Handles requests, validates input, and communicates with the ML model.
Data Preprocessing	Converts categorical values, scales numerical features, and prepares input data.
Machine Learning Model	Uses the trained Random Forest model to predict loan eligibility.

Component	Description
Prediction Engine	Generates the final prediction based on processed input.
Result Module	Displays whether the loan is Approved or Rejected.

4. System Workflow

1. User opens the Smart Lender web application.
2. Applicant enters loan-related details.
3. Flask validates the input.
4. Input data is preprocessed (encoding and scaling).
5. The processed data is sent to the trained Random Forest model.
6. The model predicts loan eligibility.
7. The prediction result is displayed to the user.

5. Technologies Used

Layer	Technology
Frontend	HTML5, CSS3
Backend	Flask (Python)
Machine Learning	Scikit-learn
Data Processing	Pandas, NumPy
Model Storage	Pickle (.pkl)
Version Control	Git & GitHub
IDE	Visual Studio Code

6. Advantages of the Architecture

- Modular and easy to maintain.
- Fast prediction with low response time.
- Accurate machine learning-based decision support.
- Scalable for future enhancements.
- Easy integration with banking systems.

7. Conclusion

The proposed architecture integrates a Flask web application with a trained Machine Learning model to provide a fast, reliable, and user-friendly loan eligibility prediction system. Its modular design ensures maintainability and supports future improvements such as cloud deployment, API integration, and enhanced prediction models.